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A holistic exploration of student attitudes toward AI use in higher education: an international comparison

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ABSTRACT

This study investigates higher education students' perspectives on artificial intelligence (AI) technologies and their use in learning and academic contexts. As AI increasingly influences higher education, understanding students' attitudes and ethical awareness is essential. The research involved 920 students from higher education institutions in Czechia (CZ), Turkey (TR), and Slovakia (SK), allowing for a cross-national comparison of attitudes, perceived benefits and challenges, and ethical considerations. A mixed-methods design was applied, combining quantitative survey data with qualitative responses, analyzed using descriptive and thematic approaches. Results indicate that most students have used AI technologies, more frequently in CZ and SK than in TR. The most common use was to enhance general understanding of academic topics. Students identified accessibility as the main benefit of AI, while key concerns included AI bias in CZ and AI misuse in TR and SK. Most participants perceived limited ethical awareness among students and emphasized the need for institutional guidelines to promote responsible AI use. Students also viewed teachers as moderately or less willing to integrate AI into teaching and learning.

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1. Introduction

Recent advancements in machine learning have contributed to the development of more innovative technologies for digital content creation, such as 'generative AI' (GenAI) (Hu, 2021). GenAI models employ sophisticated algorithms to develop new content, such as texts, images, audio and code. Their capacity to interpret intricate prompts and deliver outputs that resemble human work has spurred research and interest in incorporating GenAI across various sectors, especially in higher education (Chan, 2023). Their potential to bring innovations in teaching and learning in higher education sector has sparked considerable discussion (Adiguzel et al., 2023; Baidoo-Anu & Owusu-Ansah, 2023). For instance, recent studies highlight the transformative potential of ChatGPT in promoting learning and pedagogical innovation (Asio & Soriano, 2025).

AI's remarkable capacity to conduct complex tasks in education has emerged a lively debate among educators, whose perspectives on the future of teaching and learning differ widely. While some educators regard AI as the future of learning, others perceive it as a threat that could ultimately disrupt or even replace traditional educational activities. Although this debate implicitly highlights uncertainties surrounding AI's impact on education (Holmes et al., 2021), discussions suggest that it is utilised across many fields of study to support teaching, learning and assessment. However, it also brings up concerns surrounding pedagogy and ethics (Adiguzel et al., 2023; Skavronskaya et al., 2023). This study engages with this debate from the perspective of students by revealing their attitudes related to the use of AI in university studies and the associated perceived benefits, risks and ethical implications.

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While evidence reveals that GenAI use contributes significant inroads in higher education (Baidoo-Anu & Owusu-Ansah, 2023; Kasneci et al., Keegan, 2023) and its presence has substantially been enhanced (Ngo, 2023; Sullivan et al., 2023), there is still a need for growing knowledge on how university students perceive its use along with underlying ethical implications and institutional policies, particularly across various national contexts (Bearman et al., 2023; Bond et al., 2024; Chan & Hu, 2023; Ullrich et al., 2022; Wiese et al., 2025). Although previous research frequently emphasises the technical and pedagogical aspects associated with AI use in higher education, fewer studies have explored students' subjective experiences along with their impressions for ethical issues and institutional regulations governing AI usage (Bond et al., 2024; Dobrovská et al., 2025; Kamali et al., 2024). Moreover, limited attention has been given to how cultural norms, institutional maturity and national-level policies influence students' acceptance and expectations of AI technologies in academic settings (Ertmer and Ottenbreit-Leftwich, 2010; Straub, 1994). Cross-cultural studies in educational technology adoption suggest that values, such as uncertainty avoidance, power distance and attitudes towards innovation significantly affect user perceptions and behaviour (Hofstede, 2001; Lee et al., 2003).

There is a strong need for further research to depict the use of AI and the need for focused policy and regulations related to AI use in higher education (Chan & Hu, 2023; Schiff, 2022). Cross-national studies that examine how cultural and regulatory variations influence attitudes toward AI in educational settings are also notably lacking. Unlike previous studies that often focus on students' general attitudes towards AI within a single cultural context, this study addresses these gaps by enabling a holistic approach to students' attitudes towards multifaceted aspects of AI use in line with ethical aspects and related institutional regulations in three different national contexts – Czechia (CZ), Slovakia (SK) and Turkey (TR).

In selection of these nations, we have considered both theoretical and practical factors. From a theoretical perspective, we sought to incorporate a wide range of institutional and national circumstances in order to provide a more comprehensive understanding of student attitudes. More precisely, these nations were chosen for two primary theoretical reasons: First reason was their *national diversity*. Although SK and CZ, both Central European nations who are EU members, have a same history (as former Czechoslovakia), their post-independence developments in higher education legislation and digital transformation initiatives have diverged. TR, on the other hand, is not an EU member and has a unique context located between Europe and Asia. The second reason had to do with *variations in AI policy development phases* (Czech Republic AI Strategy Report, 2019; Slovakia AI Strategy Report, 2019a, 2019b; Turkey AI Ethics Guide, 2024). The higher education institutions (HEIs) in SK and TR are in relatively earlier phases of policy development process, although certain Czech HEIs have started to develop institutional frameworks for the use of AI. The chosen institution in CZ, for example, have been implementing detailed formal frameworks governing AI use in academic contexts, both for students and teachers for various stages of learning and teaching. Furthermore, *pragmatic factors* were also considered in the selection of these nations. Strong data collecting across universities was made possible by the study team's access to student populations in all three nations through pre-existing academic connections. They sought to take advantage of any potentially significant differences in these nations' institutional preparedness and policy development for the application of GenAI in education.

This comparative approach enables the identification of how differing institutional policies and cultural values may shape students' perceived opportunities and concerns regarding AI in education.

This diversity enables a comparative analysis of both similarities and distinctions that shape students' perspectives on the GenAI use in higher education through an international lens. A comprehensive overview of their attitudes will enable multifaceted interpretations related to AI use in university studies.

To summarise, this study fills an important research gap by exploring how cultural, ethical and institutional factors influence students' perceptions of GenAI in diverse national settings. The study's theoretical contribution lies in advancing understanding of cross-cultural variation in educational technology adoption and offering empirical insights into how institutional readiness and ethical frameworks interact with student attitudes.

Considering this justification and the research gap, this study aims to examine and compare students' perspectives on the use of AI in higher education, focusing on their familiarity with its potential, perceived benefits and challenges, ethical considerations and perceptions of their teachers' attitudes, while accounting for cultural and geographic differences. This multi-faceted and cross-national focus enables a more holistic understanding of students' attitudes towards AI in higher education.

2. Literature review

Since its inception, AI has been regarded as a valuable tool for driving innovative approaches in education and educational research. The integration of AI in education has created new opportunities, possibilities and difficulties for educational innovation, including the transition to customised learning and the changing role of educators (Baker et al., 2019; Holmes et al., 2018). It facilitates numerous approaches that might not be realised through traditional methods of education (Hwang & Chien, 2022). Although it has not yet been precisely assessed, AI is believed to have a strong potential to contribute teaching and learning, provided that the employment of such technologies aligns well with pedagogical, psychological and ethical theories.

This study is theoretically grounded in the Technology Acceptance Model (TAM) (Davis, 1989) and its later extensions such as the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), which explain how users come to accept and use new technologies. These models emphasise key factors such as perceived usefulness, perceived ease of use, social influence and facilitating conditions – constructs that are especially relevant for understanding students' and teachers attitudes towards AI in higher education.

In addition, the cross-national dimension of this study is informed by the assumption that national policy contexts and institutional preparedness can significantly influence both the adoption of educational technologies and individual user attitudes (Ertmer & Ottenbreit-Leftwich, 2010; Kozma, 2005).

Related to the specific scope of this current study, recent literature related to artificial intelligence in education (AIEd) explores below issues:

- a. The potential advantages and drawbacks of AIEd for students.
- b. The potential advantages and drawbacks of AIEd for teachers.
- c. The ethical aspects of AIEd.

Furthermore, differences in national educational and digital policies provide important background for interpreting student attitudes towards AI. CZ has adopted a national AI strategy since 2019 that includes education as a core component and encourages universities to develop institutional guidelines (Czech Republic AI Strategy Report, 2019). SK, although sharing a similar strategic document (Slovakia AI Strategy Report, 2019a, 2019b), has lagged behind in implementation within the higher education sector. TR has also launched recent efforts towards developing ethical frameworks for AI (Turkey AI Ethics Guide, 2024), but concrete policies in universities remain underdeveloped.

These differences in national policy implementation and institutional readiness are crucial for understanding how students in each country perceive the use, risks and ethical implications of AI in university studies. They also provide a relevant lens through which to analyse varying levels of student awareness, trust and engagement with GenAI technologies.

2.1. The potential advantages and drawbacks of AIEd for students

AI offers extensive and adaptable opportunities in education that can be practically applied across multiple areas. Recent systematic review studies suggest that the advantages of AIEd include various aspects, from delivering educational content to assessing learning outcomes, as well as many other facets of educational process (Cojean et al. 2023; Saputra et al., 2023; von Garrel & Mayer, 2023).

2.1.1. Adapting tasks on individual competences

One emerging advantage of AIEd is to design and deliver tasks based on learners' individual competences. AI systems personalise tasks for student learning (Chiu et al. 2023), enabling that learners could proceed based on their own advancement. They could process more data compared to a human being, thus could provide an opportunity for individual learning (Salem, 2011). Furthermore, they support the personalised needs of university students by offering tailored learning resources that align closely with their preferences and incorporating reflection processes to enhance their learning progress (Ouyang & Jiao, 2021). However, because student learning tasks in AI systems are often pre-designed rather than dynamically generated, they may not fully address individual needs (Munawar et al., 2018; Yang & Shulruf, 2019). Consequently, the insufficiency of adaptable learning resources stands out as a significant challenge in tailoring tasks to match each student's competence level (Chiu et al., 2023).

2.1.2. Feedback mechanism for student works

AI is also frequently used to provide students with prompt feedback. Its notable advantage in delivering feedback lies in its ability to provide machine-supported queries and immediate responses, accessible anytime and anywhere (Chiu et al., 2023). AI systems hold significant potential as a feedback mechanism, particularly in formative assessment, where the focus is on monitoring student progress and delivering continuous feedback to help students enhance their understanding (Gallardo, 2021). In line with this, AI-driven, data-informed feedback systems have the potential to enhance students' learning outcomes, self-regulation and motivation (Afzaal et al., 2021). However, the implementation of AI for delivering tailored feedback still requires enhancement, considering that the provided feedback may not address each student's unique needs. Both teachers and students would prefer a more intuitive and effective system that offers insightful, personalised guidance rather than repetitive, generic responses (Holstein et al., 2019). This aspect stands out as a remarkable drawback for the use of AI systems in the provision of feedback to student work. According to Ouyang and Jiao (2021), turning this limitation into a strength relies on the AI system's capacity for ongoing collection and analysis of learner-generated data, providing students with real-time, interactive opportunities to make informed choices about their learning paths.

2.1.3. Sophisticated human-machine conversations

An additional benefit of employing AI in higher education is its ability to foster interaction between users and systems and provide effective responses to inquiries (Bubaš & Čizmešija, 2023). The mechanism behind this interaction is that AI mimic human thought through frameworks that encompass the information and expertise of humans. The chatbots and language learning tools developed based on this method is used to support student's competences by continuous interaction (Kim et al., 2021, 2022; Kim & Kim, 2022). It is a beneficial technique that enables a user-friendly digital communication for learners. This interaction serves as a personalised tutor by providing learners with immediate assistance, answering specific questions, offering detailed explanations and recommending additional resources for further learning, even supporting students with their homework (Labadze et al., 2023; Lee, 2023). It could also contribute to learner motivation (Okonkwo & Ade-Ibijola, 2021). However, human-machine conversation is debated due to its drawbacks. First, it is because the results of this conversion might be ambiguous and could result in learning of erroneous information (Kooli, 2023; Wang et al., 2023). Second, this conversation is not produced by an academic author (Lee, 2023). Besides, it is unclear to what extent it truly enhances student learning and engagement (Chew & Chua, 2020). Therefore, human-machine conversions should be treated with caution by their users.

2.2. The potential advantages and drawbacks of AIEd for teachers

The integration of AI in educational environments has sparked both adoption and concern within the academic community, leading to an examination of its potential advantages and drawbacks (Gammoh, 2024). This integration assigns specific roles to teachers and the teaching process, while also raising concerns about its impact on teaching and learning.

2.2.1. Providing new opportunities for teachers to enhance teaching and learning

AIEd technologies offer a novel approach for teachers, modernising the process of teaching and instruction as well as classroom management. They provide educators new tools to help them teach, such as chat, learning analytics, automated communication and individualised support, improving their teaching efficiency (Ngo, 2023). These tools could assist educators in managing their classes efficiently (Niemi, 2024). More specifically, teachers could benefit from AI-assisted chat to follow up with students, answer questions, look up students' subject-specific histories and grade homework using chatbots (Aleedy et al., 2022). Chatbots can also function as virtual teaching assistants, posing questions to students based on straightforward instructions and offering guidance through a variety of inquiries (Smutny & Schreiberova, 2020). Through task automation, AI helps teachers create educational materials that meet the needs of their students and customise their learning (Ngo, 2023). Additionally, task automation diminishes educators' repetitive and laborious administrative responsibilities, including assignment grading, attendance tracking, and student progress monitoring. This allocates additional time for educators to concentrate on developing tailored and adaptive teaching for their pupils (Holstein et al., 2019; Wang et al., 2021). The use of such novel tools also supports teachers in efficiently assign and distribute learning materials, considering the specific needs of students (Chiu et al. 2023; Ngo, 2023). However, many teachers express concerns about their limited understanding of how these technologies work. This could lead teacher to experience feeling a loss of control, which can lower their self-efficacy and potentially discourage them from using AI to enhance their classroom teaching (Chiu et al. 2023).

2.2.2. Offering AI-enabled learning and teaching environments

AI-enabled learning environments can personalise educational content and instructional methods to meet the unique needs of each student. By analysing data on student performance, learning speed and preferences, these systems create tailored learning experiences (Delello et al., 2024). Teachers could utilise these systems to adapt instructional content into the needs and readiness of target groups. Notable examples of AI-enabled teaching and learning environments involve adaptive learning systems, intelligent tutoring systems and recommender systems (Kabudi et al., 2021). Adaptive learning systems are novel systems that could be utilised by teachers to facilitate student learning. Their function is to adjust task difficulty based on student performance (Chiu et al. 2023). Belonging this umbrella term, intelligence tutoring systems employs AI techniques to emulate a human tutor, enhancing learning by offering improved support tailored to the learner's needs (Corbett et al., 1997; Graf & Kinshuk, 2014). Recommender systems that use software tools offer suggestions for potentially useful items to learners' interest based on machine learning and information retrieval techniques (Syed et al., 2017). These solutions provide novel teaching strategies to teachers and help them to retain high-quality teaching and learning while also enhancing learning design. However, these systems reveal some drawbacks. First, there is a shortage of practical testing for these intelligent systems (Chiu et al. 2023). The majority of such systems are in the experimental phase and they are not yet commonly deployed in actual educational contexts. Secondly, there is an absence of established criteria for evaluating the effectiveness of these systems (Weragama & Reye, 2014).

2.3. The ethical aspects of AIEd

The ethical aspects related to AIEd are a matter of ongoing debate among scholars. Even though AI has the potential to modernise teaching and learning in higher education (Holmes et al., 2019), it reveals several risks that should be considered by learners, teachers and decision makers in education. One prominent risk concerns *data privacy and protection*. This is because AI relies on machine learning, which requires large data sets for its training, and such data sets pose severe risks to data protection (Carsten Stahl, 2021). Such concerns over data privacy protection, consent protocols, data breach risks, anonymity and confidentiality, cybersecurity measures and transparency in data handling were highlighted by Aguilar (2023), who noted, 'We are navigating a minefield of data privacy issues with AI' as some teachers said. The potential risk of data privacy and protection might result in breaches of student data or more that would hamper the potential benefits of AI in teaching and learning. It gradually becomes a much more serious concern because the more AI penetrates into education and training at various

levels, the more data is collected within educational contexts, and the more it serves the companies engaged in surveillance capitalism (du Boulay, 2022; Williamson, 2018).

Another significant debate related to AI is the issue of *bias*. The risk of algorithmic bias and the importance of ensuring equity are key concerns. Significant sources of AI bias fall mainly into three categories: computational bias, human bias and systemic institutional and societal bias (Schwartz et al., 2022). While considerable efforts have been made to address computational and statistical bias, remarkably less attention has been given to human and systemic bias, and such biases can cause AI to discriminate based on gender, race or socioeconomic status (Roshanaei, 2024). More critically, AI may unintentionally or even intentionally reproduce existing biases (Carsten Stahl, 2021). This means that bias produced and reproduced in AI-integrated educational settings might have undesirable outcomes during teaching and learning. To maximise the benefit from AI in education, AI training should be unbiased and AI systems should be regularly monitored to ensure ethical standards. In addition, Lee et al. (2023) pointed out the necessity of informing students about AI's technical limitations and ethical implications.

AI misuse is another common ethical concern in university studies. The rapid development of AI tools provides a new mechanism for academic misconduct which can hardly be detected or proved (Abd-Elal et al., 2019). As Keegan (2023) remarked, an AI tool has the potential to be used as a 'plagiarism machine' by students. It has multifaceted negative consequences. First, this undermines students' developing competencies for independent research and critical or analytical thinking (Jones, 2023). Second, it leads to unfairness in the assessment of student outcomes, giving some students an unfair advantage over others (Cotton et al., 2024). AI challenges the way universities evaluate the learning outcomes of students, as it is used by many students to generate any written work, including assignments and theses (Ivanov, Jones, 2023). Third, it is a predictor of unproductive behaviour in the future workplace (Guerrero-Dib et al., 2020). However, academic fraud itself could be considered as a case independent from AI. The reason behind this statement is explicit: students who misuse AI are likely to cheat either with or without using AI to do so (Rogers et al. 2024). Nonetheless, HEIs have the potential to revisit their policies and strategies to eliminate the negative aspects of AI misuse and the resulting impact on students and the quality of university studies.

Institutional AI policies are crucial in influencing students' knowledge, actions, and attitudes in the current debates about the ethical use of artificial intelligence (AI) in higher education. Only a limited number of institutions have established guidelines and frameworks for the ethical and responsible deployment of AI technologies (Delello et al., 2025; Shaw et al., 2023). A UNESCO survey encompassing 450 educational institutions worldwide reveals that less than 10% have specific guidelines and/or institutional frameworks in place regarding the usage of AI applications (UNESCO, 2023). CZ have started to formally establish university-level frameworks on AI usage, defining appropriate, partially appropriate, and inappropriate applications of AI technologies for both students and teachers (Dobrovská et al., 2025). The chosen Czech institution, for example, has established extensive guidelines outlining permissible use of AI for tasks like content editing, literature searches and grammatical correction, but restricting its usage in projects that call for creative analysis or reasoning. The Slovak institution that was part of this study, on the other hand, has not yet established AI-specific norms, so students there are expected to depend on more general rules about academic integrity. Similar to this, the Turkish university encourages the ethical and responsible use of AI, but at the time of data collection, it lacked comprehensive institutional norms. Ally and Mishra (2025) point out that there is a strong need for institutional AI policies and students should be informed about these policies and guidelines when deploying these technologies. Considering institutional clarity – or lack thereof – can greatly impact views of what constitutes ethical or permitted use of GenAI technologies in academic contexts, these differing levels of institutional policy development could provide a useful lens through which to examine student opinions.

3. Research description and research questions

Students' attitudes towards AI can vary widely depending on national and institutional factors, necessitating a comprehensive analysis that spans multiple countries and educational systems. Deploying a

cross-national lens, this study aims to examine and compare students' perspectives on the use of AI in higher education, focusing on their familiarity with its potential, perceived benefits and challenges, ethical considerations and perceptions of their teachers' attitudes. Its multi-faceted and cross-national aspect enables a more holistic understanding of students' attitudes towards AI use in higher education. This research will specifically delve into varying attitudes of students towards AI use across three countries: CZ, TR and SK. The following research questions form the base of this study:

RQ1: How familiar are the students with the potential of AI technologies?

RQ2: What are the perceived pros and cons associated with using AI in university studies, as seen by students?

RQ3: How do students consider ethical aspects of AI use and see universities' setting guidelines for AI use?

RQ4: How do students view their teachers' attitudes towards AI use in higher education?

RQ5: Are there any differences in the attitudes of students in the selected countries?

4. Research methodology

A mixed-methods approach was employed to capture both quantitative and qualitative dimensions of AI use among students. The quantitative component involved a structured survey, while the qualitative component comprised open-ended questions analysed thematically. Cultural factors, including national dimensions (Hofstede, 2011) and technology acceptance frameworks (UTAUT), were considered to anticipate potential differences in AI adoption and perceptions among students from different countries. The design was selected for its effectiveness in capturing broad patterns and quantifiable insights across different national contexts, specifically among students from the Czech Republic, TR and SK. To enrich the findings, the study incorporates both quantitative and qualitative research approaches to allow for both statistical analysis and for nuanced perspectives.

The selection of these three countries was based on a combination of theoretical and practical considerations as described in the introductory part of this text. Theoretically, the countries represent diverse stages of digital transformation and institutional readiness in higher education, as well as cultural and regional variation within Europe and its periphery. CZ and SK share a common historical foundation as former Czechoslovakia but have developed distinct educational systems and policy responses to AI since their separation. TR, as a non-EU country situated at the intersection of Europe and Asia, offers a contrasting context in terms of digital policy, educational governance and societal attitudes towards technology (Czech Republic AI Strategy Report, 2019; Slovakia AI Strategy Report, 2019b; Turkey AI Ethics Guide, 2024). Practically, academic collaborations with partner universities in all three countries enabled access to student populations and ensured smooth administration of the data collection process.

4.1. Participants

The study's participants include a total of 920 (CZ $n = 302$, TR $n = 311$, SK $n = 307$) undergraduate students enrolled in full-time programmes at selected universities in each country from Arts and Humanities and Education Faculties in CZ, TR and SK. Exclusion criteria included incomplete surveys or refusal to provide consent. A convenience sampling technique was utilised to select participants based on their availability and desire to engage in the study. The questionnaire was distributed online via email, social media platforms and educational institution networks to reach a broader audience. Participants were briefed on the purpose of the study and informed consent was obtained before they completed the survey. Table 1 shows the characteristics of the participants.

As seen in Table 1, the study group consists of participants from three countries: CZ ($n = 302$), TR ($n = 311$) and SK ($n = 307$). A variety of students with different characteristics (gender, grade level, age and faculty) were involved in the research. To confirm that deviations in the distribution of characteristics of the study sample may be attributed to random variation, we provided an example of assessing the significance of the difference in the representation of male and female participants in the study.

Table 1. Characteristics of participants from Czechia, Turkey and Slovakia.

Characteristics		Czechia (<i>n</i> = 302)		Turkey (<i>n</i> = 311)		Slovakia (<i>n</i> = 307)	
		<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
Gender	Male	131	43.4	107	34.4	76	24.8
	Female	161	53.3	198	63.7	225	73.3
	Non-defined	10	3.3	6	1.9	6	1.9
Grade Level	1	106	35.1	41	13.2	131	42.7
	2	104	34.4	123	39.5	129	42.0
	3	79	26.2	70	22.5	31	10.1
	4	13	4.3	77	24.8	16	5.2
Age	19 or less	5	1.7	31	10.0	27	8.8
	20	26	8.6	78	25.1	72	23.5
	21	23	7.6	61	19.6	67	21.8
	22	31	10.3	69	22.2	55	17.9
	23	25	8.3	43	13.8	36	11.7
	24	19	6.3	12	3.9	21	6.8
	25–30	47	15.6	17	5.5	18	5.9
	31 or more	126	41.7	–	–	11	3.6

Table 2. Chi-square test for association by countries.

	Male (observed)	Male (expected)	Female (observed)	Female (expected)
Outcome 1: Czechia	131	102	161	190
Outcome 2: Turkey	107	107	198	198
Outcome 3: Slovakia	76	105	225	196
Total	314		584	

Table 2 shows the respondents' observed values and the corresponding expected values. We always determined the expected values for the partially combined frequency of sorting by gender and country (Czech Republic = Outcome 1, TR = Outcome 2 and SK = Outcome 3). And we calculated this expected frequency using the intersection of marginal probabilities (the product of the respective marginal frequencies to the pooled frequency divided by the total frequency of the set [$N = 898$]). Subsequently, we determined the chi-squared sum of the relative squared deviations between the observed and expected values according to the relationship:

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

After substituting the values from the table into the previous relationship, we get the value of the test criterion.

$$\chi^2 = (131 - 102)^2 / 102 + (107 - 107)^2 / 107 + \dots + (225 - 196)^2 / 196 = 24.971 > 5.991 = \chi_{0.05(2)}^2$$

where $\chi_{0.05(2)}^2 = 5.991$, the critical value for $\alpha = 0.05$ significance level and 2 degrees of freedom is ($2 = (3 - 1) \times (2 - 1)$). Since the calculated value of the test criterion is greater than the critical value, we can reject the null hypothesis of equality in the arrangement of respondents by gender at the 0.05 significance level. This reveals that the deviations in the distribution of characteristics of the study sample could be due to mere chance.

4.2. Data collection tool and its administration

A questionnaire developed by Dobrovská et al. (2024) was adapted and administrated within the frame of this study. Data were gathered from June to October 2024. The data collection lasted approximately 12 weeks. The body of original questionnaire was comprised 12 items including close-ended and open-ended questions. During the design of the original questionnaire, the relevant literature was revisited, several students were interviewed to gain their insights into the issue, and expert opinions were collected to draft the questionnaire. The final form was defined after modifications to the items based on expert opinions and pilot implementation phase (Dobrovská et al., 2024). This tool was adapted considering the scope and objectives of the current study. During the adaptation process, the aim and scope

of the study was defined first. Two more items were added to the questionnaire considering the relevant literature and the scope of the current study.

4.2.1. Cross-Cultural adaptation and validity procedures

To ensure cross-cultural validity, the following procedures were implemented:

- The questionnaire was first translated into Czech, Slovak and Turkish by professional bilingual translators.
- Each translated version underwent back-translation by independent translators to verify conceptual equivalence with the original English version.
- Three subject-matter experts from each country (nine total) with expertise in AI in education and questionnaire design reviewed the translated items for clarity, relevance and cultural appropriateness.
- A pilot implementation phase was conducted in each country ($n \approx 10$ per country), during which students provided feedback on item clarity and relevance. Modifications were made accordingly to improve comprehensibility and ensure contextual appropriateness.
- Although formal measurement invariance testing (e.g. confirmatory factor analysis) was not conducted due to the exploratory and partly qualitative nature of the instrument, the above procedures were implemented to increase linguistic, semantic and functional equivalence across the three country contexts.

The final instrument comprised 14 questions divided into three parts: (1) general instructions and study purpose; (2) participant demographics; and (3) the main questionnaire on students' attitudes towards AI use.

4.3. Data analysis

Qualitative data were analysed using MAXQDA for thematic coding. Quantitative data were analysed using descriptive statistics in MS Excel (frequency and percentage), and the survey's open-ended items using a thematic analysis technique. The thematic analysis was based on Braun and Clarke's (2006) approach, including six core stages for data analysis: Familiarisation with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes and writing the report. We selected this approach because of its flexibility and its emphasis on finding themes or patterns in qualitative data. Because of its thorough framework for evaluating big datasets, similar to the current study's dataset, this approach is frequently used in qualitative research. In the process of thematic analysis, three authors first reviewed the replies iteratively and coded different parts of them to generate themes. The coding procedures and choices on groups and themes were discussed in several correspondences between the three authors. To ensure consistency, we assessed inter-rater reliability using Cohen's kappa, the result of which is a coefficient of $\kappa = 0.82$, a high-level consistency amongst authors. Nonetheless, we held further meetings to build consensus over incompatible codes and themes. The overall procedure served for reliability of the data. As for its validity, the study group was identified in detail, and procedures of questionnaire development, data collection and analysis processes were reported in detail.

5. Research results from Czechia, Turkey and Slovakia

5.1. Students' familiarity with the potential of AI technologies

This research elucidates student attitudes towards AI technologies and their use in higher education. In this context, the research first reveals how familiar the students are with the potential of AI technologies. Table 3 indicates the prevalence and purpose of AI use among university students.

The results reveal that most of the participants from all three countries have used AI technologies at least once, while higher usage rates observed in CZ (82.1%) and SK (81.1%) compared to TR (60.5%). Notably, the majority of participants who reported using AI technologies described their frequency of use as 'sometimes' or 'commonly'. The most common purpose of using AI among the participants from all three countries is to have a general idea of the topic. This suggests that students view AI primarily as a tool for orientation in new content rather than as a substitute for deeper learning. Remarkably, many

Table 3. Prevalence and purpose of AI use.

		Czechia (n = 302)		Turkey (n = 311)		Slovakia (n = 307)	
		n	%	n	%	n	%
Use of AI technology	Yes	248	82.1	188	60.5	249	81.1
	No	54	17.9	123	39.5	58	18.9
Frequency of use	Nearly not at all	17	6.9	6	3.2	14	5.6
	Rarely	58	23.4	59	31.4	68	27.3
	Sometimes	105	42.3	85	45.2	93	37.3
	Commonly	62	25.0	28	14.9	56	22.5
	Always	6	2.4	10	5.3	17	6.8
Purpose of using AI	Gain a general understanding of a topic	185	74.6	128	68.1	174	69.9
	Process information	139	56.0	75	39.9	173	69.5
	Do an assignment	69	27.8	114	60.6	47	18.9
	Translate a text	92	37.1	55	29.3	85	34.1
	Prepare research articles	27	10.9	31	16.5	20	8.0

Table 4. AI tools used by university students.

Czechia			Turkey			Slovakia		
AI tool	n	%	AI tool	n	%	AI tool	n	%
ChatGPT	185	74.6	ChatGPT	149	79.3	ChatGPT	146	58.6
DeepL	18	7.3	Kahoot! AI Assistant	16	8.5	DeepL	6	2.4
Copilot (Bing)	15	6.0	Canva AI Assistant	13	6.9	Copilot (Bing)	6	2.4
Gemini (Bard)	10	4.0	DeepL	7	3.7	ChatPDF	6	2.4
DALL-E	9	3.6	Copilot (Bing)	6	3.2	Photomath	6	2.4
ChatBot	6	2.4	Wordwall AI Assistant	5	2.7	Grammarly	5	2.0
Deeply	6	2.4	Bard AI	3	1.6	ChatBot	4	1.6
Midjourney	4	1.6	DALL-E	3	1.6	Gemini (Bard)	4	1.6
Perplexity	3	1.2	Gemini (Bard)	3	1.6	DALL-E	3	1.2
Adobe Podcast AI	2	0.8	ChatBot	2	1.1	GeoGebra	3	1.2
Deep AI	2	0.8	Quizlet AI Assistant	2	1.1	Perplexity	3	1.2
DesignsAI	2	0.8	Smoi AI	2	1.1	QuillBot AI	2	0.8
			Snapchat My AI	2	1.1	Siri	2	0.8

*AI tools used by at least 2 students are available in the table.

of the participants declared having used AI to do assignments and prepare research articles. Based on the first research question on how familiar the students are with the potential AI technologies; [Table 4](#) shows the AI tools that students use in their university studies.

The results highlight that the majority of the participants from all three countries employ ChatGPT as an AI tool in their university studies. Strikingly, students in the three countries employ a variety of AI tools, each tailored to specific purposes in their university studies.

5.2. Pros and cons of using AI in university studies

The research then investigates the pros and cons associated with using AI in university studies, based on the opinions of students. [Table 5](#) indicates the differences and similarities among the views of participants from the three different countries.

In terms of the *pros* of using AI in university studies, results highlight that students in all three countries think the most prevalent pro is 'It is easy to access, available 24/7' and the least is 'It provides a more personalized learning experience'. The similarity across countries in perceiving accessibility as the main advantage suggests a shared reliance on AI as a convenient on-demand support. When it comes to the *cons* of using AI, students in CZ emphasise algorithmic bias (72.2%), whereas Turkish and Slovak students are more concerned about misuse. While the survey presents frequency data, further analysis reveals practical implications:

- The high concern about misuse and laziness points to the necessity for universities to guide students in responsible use.
- Conversely, the low ratings of personalisation suggest current tools may lack adaptability to individual learning needs.

Table 5. Pros and cons of using AI in university studies.

		Czechia (n = 302)		Turkey (n = 311)		Slovakia (n = 307)	
		n	%	n	%	n	%
Pros of using AI	It is easy to access, available 24/7	212	70.2	229	73.6	248	80.8
	It is time saving	203	67.2	216	69.5	224	73.0
	It helps with assignments	122	40.4	205	65.9	143	46.6
	It helps with data analysis	141	46.7	159	51.1	165	53.7
	It easily creates materials	114	37.7	145	46.6	157	51.1
	It helps with decision-making	88	29.1	125	40.2	75	24.4
Cons of using AI	It provides a personalized learning experience	34	11.3	104	33.4	39	12.7
	It has the potential for misuse	167	55.3	216	69.5	232	75.6
	There is bias in its algorithms	218	72.2	90	28.9	220	71.7
	It makes students lazier	158	52.3	193	62.1	162	52.8
	There is a risk of excessive reliance on AI	153	50.7	160	51.4	187	60.9

Table 6. Students' consideration level of ethical aspects.

		Czechia (n = 302)		Turkey (n = 311)		Slovakia (n = 307)	
		n	%	n	%	n	%
Students' consideration level of ethical aspects in university studies							
Never		36	11.9	30	9.6	33	10.7
Rarely		101	33.4	126	40.5	125	40.7
Sometimes		115	38.1	120	38.6	111	36.2
Often		37	12.3	29	9.3	25	8.1
Always		13	4.3	6	1.9	13	4.2

Although the study does not apply inferential statistical tests in this section, the high percentage differences (e.g. 72.2% vs. 28.9% for algorithmic bias) indicate practically significant trends that should not be overlooked.

5.3. Ethical aspects related to AI use in university studies

The research also highlights ethical aspects of AI use in university studies, as perceived by students. In line with this, Table 6 indicates students' consideration level of ethical aspects in university studies.

According to the survey results, the majority of students in each of the countries think that ethical aspects in AI use in their university studies are considered at a rare to moderate level. The majority of students in each of the countries think that ethical aspects in AI use in their university studies are considered at a rare to moderate level. Remarkably, only 16.6% of students in CZ, 13.2% in TR and 12.3% in SK think that ethical aspects in AI use are often or always considered in their university studies. On the other hand, strikingly, 45.3% of students in CZ, 50.1% in TR and 51.4% in SK think that ethical aspects in AI use are either never or rarely considered in university studies. Complementary to the above results, the research also reveals students' opinions on whether universities should set guidelines for ethical and unethical behaviours, as indicated in Table 7. This data has not only statistical weight, but also practical significance: With 65.2% (CZ), 80.7% (TR) and 60.1% (SK) supporting the creation of guidelines, universities are clearly expected to act.

Qualitative analysis results based on open-ended questions reveal that the most common reason in all countries is the risk of AI misuse by students. However, their suggestions for university management differ among the countries. On the other hand, the ratio of students suggesting that universities should not set ethical and unethical guidelines for behaviours remains a minority opinion.

5.4. Teachers' attitudes towards AI use

Research lastly reveals students' views about their teachers' attitudes towards AI use in higher education. Table 8 shows participant students views about their teachers' attitudes.

Survey results reveal students' views about their teachers' attitudes towards AI use in three distinct pillars: teachers' belief in the potential of AI, their openness to improve their understanding of AI, and their actual use of AI tools. First, the results indicate that the majority of the students think their teachers believe AI has the potential to enhance teaching and learning at a moderate level, while only 7.3% of the students in CZ, 22.2% in TR and 5.9% in SK think that their teachers have a high or very high

Table 7. Students' opinions about universities setting guidelines for ethical and unethical behaviours.

	Czechia		Turkey		Slovakia	
	Theme	<i>f</i>	Theme	<i>f</i>	Theme	<i>f</i>
University should SET guidelines for ethical and unethical behaviours. [CZ: 65.2%, <i>f</i> = 197] [TR: 80.7%, <i>f</i> = 251] [SK: 60.1%, <i>f</i> = 212] Because:	Eliminate the risk of AI misuse by students	27	Eliminate the risk of AI misuse by students	43	Eliminate the risk of AI misuse by students	59
	Ensure 'real' learning and quality of studies	14	Encourage a controlled and mindful AI use	12	Ensure 'real' learning and quality of studies	19
	Create awareness of ethical AI use	6	Eliminate injustice among students	9	Eliminate injustice among students	12
	Ensure true and reliable information	5	Create awareness of ethical AI use	6	Lessen the overuse of / overreliance on AI	11
	Eliminate injustice among students	4	Ensure true and reliable information	1	Ensure true and reliable information	6
	Ensure transparency and accountability	2			Benefit truly from technology for studies	2
	Avoid degrading the value of diplomas	1			Protect the meaning of university studies	2
	No clear line between what is ethical and not	1			Risk getting out of control if not controlled	2
	Identify ethical and unethical AI use	53	Offer clear information on AI ethics in meetings	53	Define AI policy and set boundaries for AI use	46
	Define the ratio and domains of allowed AI use	17	Identify ethical and unethical AI use	33	Screen originality and evaluate accordingly	11
University could DO:	Screen originality and evaluate accordingly	9	Define the ratio and domains of allowed AI use	20	Change / modernize the assignment process	7
	Inform/educate students about ethical AI use	8	Screen originality and evaluate accordingly	12	Ask students to cite AI as a source	6
	Ask students to cite AI as a source	5	Offer elective courses about AI use and ethics	5	Forbid AI use for university studies	5
	Change / modernize the assignment process	3	Ask students to cite AI as a source	2	Offer clear information on AI ethics in meetings	3
	Integrate AI ethics into courses	1	Forbid AI use for university studies	5		
			Define penalties for AI misuse	2		
			Integrate AI ethics into courses	1		
	Impossible to control compliance with rules	16	Ethical justification should be set by oneself	17	AI is a good helper that supplements studies	20
	Ethical justification should be set by oneself	9	Individuals should be free in their actions	5	Impossible to control compliance with rules	16
	This would reduce the benefits of true AI use	8	Impossible to control compliance to rules	3	Ethical justification should be set by oneself	6
University should NOT SET guidelines for ethical and unethical behaviours. [CZ: 34.8%, <i>f</i> = 105] [TR: 19.3%, <i>f</i> = 60] [SK: 30.9%, <i>f</i> = 95] Because:	Assignments be redesigned considering AI	7	This would reduce the benefits of true AI Use	2	Assignments be redesigned considering AI	4
	Individuals should be free in their actions	2	Assignments be redesigned considering AI	2	Risks and harms belong to students themselves	4
	Ethical training precedes banning	1	Universities lack technical background	1	This would reduce the benefits of true AI Use	3
					Individuals should be free in their actions	2
					Educating students and teachers is enough	1
					Ethical rules are vague	1

level of belief that AI has the potential to enhance teaching and learning. Similarly, second, the majority of the students in CZ and TR think their teachers are at a moderate level open to improve their understanding of the use of AI during teaching and learning, while the majority of students in SK are at a low level. Only 6.9% of the students in CZ, 20.9% in TR and 7.5% in SK think their teachers are at a high or very high level open to improve their understanding of the use of AI during teaching and learning. On the other hand, the majority of the students in TR think their teachers have a moderate level of AI use during teaching and learning, while the students in CZ report a low level and in SK a very low level. Only 4.3% of the students in CZ, 16.4% in TR and 4.5% in SK think their teachers use AI during teaching and learning at a high or very high level. Remarkably, students from CZ and SK have a similar level of attitudes regarding their teachers' overall attitudes towards AI, while students from TR have relatively higher-level attitudes regarding their teachers' overall attitudes towards AI use in teaching and learning in higher education. Some studies show that teacher attitudes toward AI significantly influence its adoption in education (Aghaziarati et al., 2023).

These results have important educational relevance:

- They suggest a significant gap between students' use of AI and teacher adoption.
- This gap could lead to misalignment of expectations, potentially undermining efforts at academic integrity or innovative pedagogy.
- From a practical perspective, teacher training in AI literacy emerges as a key area for intervention across all countries.

6. Discussion and conclusions

This study was designed to provide a holistic examination of students' attitudes towards AI use in higher education from a cross-national perspective. This study provides insights into students' attitudes towards AI in higher education across CZ, SK and TR. The findings align with and extend prior research by highlighting cross-national differences in AI engagement and ethical awareness. Students use AI primarily for understanding concepts rather than for complex academic tasks, suggesting limitations in digital literacy or confidence. Ethical concerns (misuse, bias and overreliance) were prominent, particularly where institutional guidelines are lacking. A variety of students ($n=302$ in CZ, $n=311$ in TR, and $n=307$ in SK) with different characteristics were involved in the research, contributing to a comprehensive interpretation of the results. Both descriptive analysis (frequency and percentage) and thematic analysis approaches were conducted to address the five research questions, offering insights into student behaviour, perception and expectations related to AI in university contexts.

First, the findings reveal that while students across all three countries have engaged with AI technologies, usage frequency varies – most notably with CZ and SK demonstrating higher engagement than TR. Differences in AI use and perception across countries may be partially explained by national digital readiness indices and AI policy frameworks (Czech Republic AI Strategy Report, 2019; Maslej et al., 2024; Slovakia AI Strategy Report, 2019a, 2019b; Turkey AI Ethics Guide, 2024). Students from countries with higher digital infrastructure and clearer AI guidelines demonstrated more frequent and confident AI use. This may reflect broader digital infrastructure readiness and earlier national-level AI strategy implementation in the former two countries (Maslej et al., 2024). These contextual factors likely influence the availability and normalisation of AI tools in higher education environments.

The purposes for which students use AI tools differ slightly across the countries, but the most common application is to gain a general understanding of topics. Interestingly, despite AI's potential for deep academic utility, using AI to prepare research articles was the least common across all three nations. Although literature suggests a variety of purpose for using AI tools in education (Chen et al., 2020), this may relate to students' limited digital literacy or lack of confidence in applying AI tools for more complex academic tasks.

ChatGPT emerged as the most widely used AI tool among students in all three countries, as reported in literature (Aydin & Karaarslan, 2023). It is followed by DeepL, Copilot, and Gemini. Visual-based AI tools like Kahoot! Canva and Wordwall were notably more popular among Turkish students, likely due to the predominance of education majors in the Turkish sample. The disciplinary composition of the

Table 8. Teachers' attitudes towards AI use.

The level of teachers' ...		Czechia (<i>n</i> = 302)		Turkey (<i>n</i> = 311)		Slovakia (<i>n</i> = 307)	
		<i>n</i>	%	<i>n</i>	%	<i>n</i>	%
... believing AI has the potential to enhance teaching and learning	Very low	50	16.6	22	7.1	51	16.6
	Low	97	32.1	60	19.3	116	37.8
	Moderate	133	44.0	160	51.4	122	39.7
	High	20	6.6	61	19.6	15	4.9
	Very high	2	0.7	8	2.6	3	1.0
... being open to improve their understanding towards the use of AI during teaching and learning	Very low	67	22.2	26	8.4	74	24.1
	Low	98	32.5	57	18.3	112	36.5
	Moderate	116	38.4	163	52.4	98	31.9
	High	17	5.6	56	18.0	20	6.5
	Very high	4	1.3	9	2.9	3	1.0
... using AI tools during teaching and learning	Very low	103	34.1	42	13.5	125	40.7
	Low	131	43.4	87	28.0	103	33.6
	Moderate	55	18.2	131	42.1	65	21.2
	High	11	3.6	49	15.8	13	4.2
	Very high	2	0.7	2	0.6	1	0.3

samples also influenced AI engagement; for example, Turkish students, largely from education programmes, preferred visual AI tools (Canva, Kahoot!), suggesting that field of study shapes tool preference and perceived utility. This suggests that disciplinary orientation may shape AI tool preferences.

Regarding perceived benefits, students emphasised accessibility and time-efficiency, aligning with global trends. Participants also reported other benefits, such as being time-saving, assignment assistance, data analysis, decision-making, and material creation. This result coincided with the relevant literature (Perera & Lankathilaka, 2023; Su & Yang, 2023; Samala et al., 2024). However, the relatively low recognition of AI's capacity for personalised learning suggests that students currently use AI for task completion rather than enhancing learning processes. This may reflect limited exposure to adaptive AI systems or a focus on short-term academic outputs. This is striking because AI is reported to have a good potential to improve teaching and learning, adapting the learning process to the needs of learners (Harry & Sayudin, 2023; Pisica et al., 2023; Qadir, 2023).

The perceived disadvantages of AI were more nuanced. Students frequently cited three major concerns: the potential for misuse, algorithmic bias and overreliance on AI. The concern over misuse – particularly high in TR and SK – may be linked to the absence of formal institutional policies on AI usage in these countries, unlike CZ where clearer guidelines exist. This underlines the importance of regulatory clarity in alleviating ethical concerns.

Bias was another significant concern, particularly among students from CZ and SK, while Turkish students mentioned it less frequently. This may indicate lower exposure to critical discussions of AI fairness and ethics in TR. Educators and institutions may need to make greater efforts to integrate AI ethics into course content and co-curricular activities.

A third prominent concern was overreliance on AI, which students perceived as potentially detrimental to critical and analytical thinking (Zhai et al., 2024). These concerns reinforce the argument for developing digital literacy programmes that promote mindful and critical AI use.

Ethical considerations surfaced strongly in student responses, with many admitting they seldom reflect on the ethical dimensions of AI use. This finding is especially concerning, given that unchecked AI usage could compromise academic integrity and degrade the quality of learning. Students expressed a strong preference for institutional guidelines on ethical AI use, further validating the need for structured policies. Countries and institutions with fewer existing guidelines, such as TR and SK, showed higher student demand for regulation.

Students also expressed opinions on the attitudes of their instructors towards AI in teaching and learning. Most believed that their instructors had moderate belief in AI's potential and low to moderate openness to using it. Notably, perceived instructor beliefs and openness exceeded their actual use of AI in classrooms. This discrepancy may be rooted in ethical hesitations, lack of confidence or concerns about changing established pedagogical routines. Teachers may also view AI as a challenge to their traditional role, further slowing adoption (Aghaziarati et al., 2023; Alwaqdani, 2025). In addition, teachers might have concerns about their own potential to shift their pedagogy towards AI-based instruction (Kim, 2024; Nazaretsky et al., 2021). This is because it will require a change in their class routines and in

their management approach to students following different learning sequences. Furthermore, teachers might consider AI as a potential threat to their conventional role (Humble & Mozelius, 2022; Popenici & Kerr, 2017). Interestingly, Turkish students reported more favourable perceptions of their instructors' AI-related attitudes, possibly reflecting more informal and flexible AI practices in their institutions.

Observed gaps between student expectations and teacher readiness highlight the need for targeted faculty development, institutional support and integration of AI literacy into curricula, which may help bridge the student-teacher adoption gap. These findings point to a potential misalignment between student expectations and teacher readiness, highlighting the need for faculty development and clearer institutional support.

As seen from our results insights into students' attitudes towards AI use in higher education across CZ, TR and SK, providing a foundation for informed decision-making among stakeholders in education.

For Curriculum Developers:

- **Integrate AI Literacy into Curricula:** AI should not be treated merely as a tool for task automation but as a subject of inquiry. Educators should design interdisciplinary modules that cover AI fundamentals, ethical considerations, algorithmic bias and responsible use.
- **Foster Critical Digital Skills:** The low emphasis on personalised learning and ethical awareness among students suggests a need for curriculum components that develop students' analytical, evaluative, and ethical reasoning skills when engaging with AI tools.
- **Promote Contextual and Purposeful AI Use:** Curricula should help students differentiate between appropriate and inappropriate uses of AI for academic tasks, emphasising AI as a support for thinking, not a replacement.

For Institutional Leaders:

- **Establish Clear AI Policies and Guidelines:** Students strongly support university-level guidance on ethical and unethical uses of AI. Institutions should implement transparent AI usage policies co-created with students and faculty, specifying permissible use cases, citation norms and academic integrity standards.
- **Provide Faculty Training and Incentives:** To address the observed gap between student expectations and teacher readiness, universities should invest in continuous professional development. Training should include pedagogical strategies for incorporating AI, mitigating bias and managing AI-enhanced assessments.
- **Create Innovation Hubs:** Universities can foster responsible experimentation with AI by establishing digital learning labs or AI hubs where students and faculty co-design AI-enhanced learning experiences.

For Education Policymakers:

- **Develop National Guidelines for AI in Education:** While some countries like CZ and SK have national AI strategies, the disparity in student engagement and awareness suggests that these need more explicit integration into higher education policies. National frameworks should address AI ethics, accessibility and teacher training.
- **Invest in Infrastructure and Access:** Digital readiness still varies significantly among countries. Policymakers should prioritise equitable access to high-speed internet, AI tools and digital devices, particularly in under-resourced institutions.
- **Mandate AI Ethics Education:** Ethical understanding should be a mandatory component of digital literacy programmes at the tertiary level. Governments should fund initiatives to embed AI ethics and social impact studies into all academic disciplines.

By aligning curricula, institutional policies, and national strategies, stakeholders can create a responsible, equitable and forward-looking AI-integrated higher education landscape.

7. Conclusion

This cross-national study contributes to literature on AI in higher education by addressing student engagement, perceptions, and ethical considerations. Despite widespread AI use, students reported limited engagement with advanced features, and expressed a clear need for institutional guidance on ethical AI practices.

Students appreciate AI's practical benefits, particularly its accessibility and time-saving capabilities. However, their limited engagement with AI's more advanced and personalised learning capabilities suggests a need for broader AI literacy and education. Ethical concerns – including AI misuse, bias and over-reliance – were prominent, though they varied in salience across countries, pointing to the influence of institutional culture and national policy.

There is a clear student-driven call for universities to establish comprehensive guidelines for the ethical use of AI in academic contexts. Furthermore, students perceive a gap between their own AI usage and that of their instructors, signalling a need for targeted teacher training, policy development and curriculum reform.

Overall, the study reinforces the urgency of integrating AI ethics, literacy and responsible use into higher education systems through aligned efforts by curriculum developers, institutional leaders and policymakers. A coordinated approach can better prepare both students and educators for the increasingly AI-enhanced academic landscape.

7.1. Limitations and future research

While this study offers meaningful cross-national insights, several limitations must be acknowledged:

Methodological constraints:

- The study employed a mixed-methods approach relying primarily on self-reported survey data, which may be influenced by social desirability bias or lack of deep reflection. Thematic analysis of open responses, while informative, is interpretive and may miss nuances. Limitations include reliance on self-reported data, limited institutional representation and disciplinary skew.

Sampling and representation:

- Although the total sample ($n = 920$) is sizable, the participants were drawn from specific universities in each country, limiting the diversity of institutional contexts represented. Students from other institutions, programmes or geographic regions may have different experiences and attitudes towards AI.
- The disciplinary skew – such as a higher proportion of Turkish students in pedagogical fields – may have influenced the types of AI tools used and perceptions of educational utility, reducing the generalisability of findings to other disciplines.

Generalisability:

- The study focuses on three Central and Eastern European countries. While this provides important regional insights, the findings may not be transferable to countries with vastly different technological infrastructure, AI policy maturity or educational cultures.

Suggestions for Future Research:

1. Include Educator Perspectives: Future studies should incorporate interviews or surveys with teachers to triangulate student-reported perceptions of faculty attitudes and behaviours regarding AI, broader geographic representation, longitudinal designs and assessments of learning outcomes.
2. Expand to Diverse Regions and Institutions: Replicating the study in different countries, including those in the Global South or high-income Western nations, would provide a more global view of student engagement with AI.

3. Longitudinal Design: A follow-up study tracking the evolution of students' attitudes and use of AI over time would provide deeper insight into changing trends, especially as national policies and tools evolve.
4. Examine Learning Outcomes: Future research should investigate whether and how AI use contributes to or detracts from student learning outcomes, critical thinking and ethical reasoning skills.
5. Explore AI Literacy and Ethics Curriculum Impact: Experimental or quasi-experimental studies can assess the effectiveness of AI ethics and digital literacy interventions in shaping student behaviour and awareness.

By acknowledging these limitations and directing future inquiry towards these areas, researchers and practitioners can deepen the understanding of AI's evolving role in higher education and design more targeted, inclusive and ethical AI education strategies.

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Availability of data and material

The datasets used and/or analysed in this study are available from the corresponding author upon reasonable request.

Disclosure statement

The authors declare that they have no conflicts of interest.

Consent to publish

Participants were informed that the results of the research would be published in an anonymised and aggregated form.

Authors' contributions

David Vaněček contributed to the conceptualisation, design, methodology, data collection, analysis, writing original draft and supervision. Yilmaz Ilker Yorulmaz contributed to conceptualisation, design, methodology, data collection, analysis, writing original draft. Dana Dobrovská contributed to the study design, writing theoretical framework, data analysis, interpretation. All authors read and approved the final manuscript.

Ethics approval

This study received ethical approval from the Ethics Committee of the Masaryk Institute of Advanced Studies, MUVS No. 01/2024/003. All research was carried out following the ethical standards and guidelines associated with research involving human participants.

Informed consent

Informed consent was obtained from all individual participants included in the study.

Statement regarding research involving human participants and/or animals

All procedures involving human participants were performed by the ethical standards of the Masaryk Institute of Advanced Studies, Czech Technical University in Prague, Czech Republic. All research was carried out following the ethical standards and guidelines associated with research involving human participants. Furthermore, there were no animals involved in the research.

Consent to participate

All participants provided consent to participate in this study.

Consent to publish

Participants consented to the publication of anonymised data.

Author contributions

DV: Conceptualisation, design, data collection, analysis, interpretation, synthesis, writing original draft and supervision.

YIY: Conceptualisation, design, data collection, analysis, interpretation, and writing original draft.

DD: Design, writing theoretical framework, data analysis, interpretation, editing.

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